Mixture Models: An Intuitive Introduction

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What are we learning here?

By the end of this material, you should be able to:

- explain the **generative process** behind a mixture model;
- understand why mixture models are useful when a population contains several underlying subpopulations;
- explain the role of the **latent variable** Z_i;
- interpret the **mixture proportions** $\pi_1, \ldots, \pi_K$;
- write down the marginal distribution of an observation from a mixture model; and
- compute the probability that an observation belongs to each mixture component.

The main idea is simple:

> A mixture model assumes that the population is made up of several underlying subpopulations. Each observation comes from one of these subpopulations, but we may not know which one.

Building the intuition first

Suppose we have two underlying groups

Imagine that observations can arise from one of two different populations. Within each population there is variability, so we model each group using its own probability distribution.

For illustration, suppose

$$X \mid Z = 1 \sim N(3, 1)$$

and

$$X \mid Z = 2 \sim N(8, 1).$$

Here Z tells us which population generated the observation.

```
set.seed(30027)

x1 <- rnorm(20, mean = 3, sd = 1)
x2 <- rnorm(20, mean = 8, sd = 1)

two_groups <- data.frame(
  x = c(x1, x2),
```

```
  component = factor(
    c(rep("Component 1", length(x1)),
      rep("Component 2", length(x2)))
  )
)

ggplot(two_groups, aes(x = x, y = component, colour = component)) +
  geom_jitter(height = 0.08, width = 0, size = 2.5) +
  labs(
    x = "Observed value",
    y = NULL,
    colour = NULL
  ) +
  theme_bw() +
  theme(legend.position = "none")
```

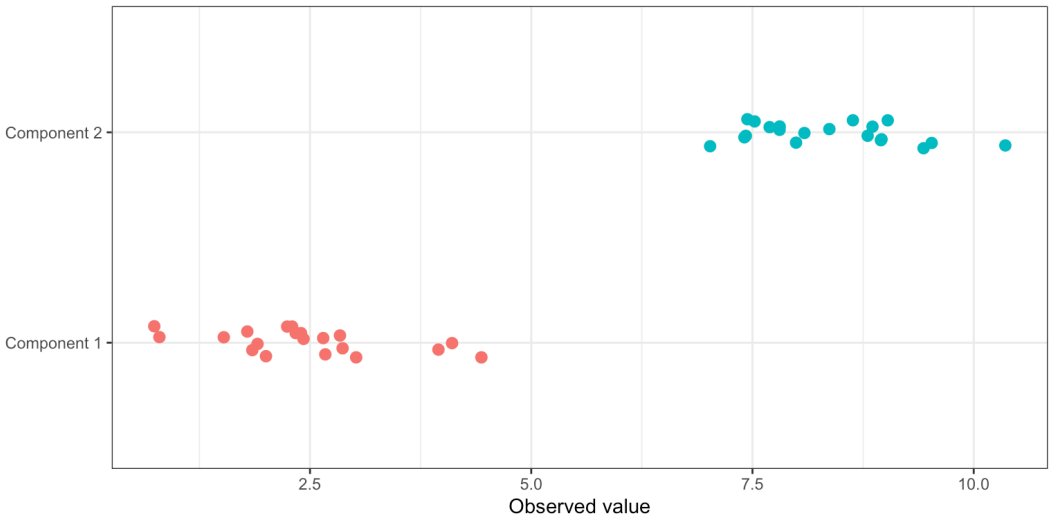


Figure 1: Observations generated from two different underlying populations.

If we knew the group labels, we could simply analyse the two groups separately.

The interesting situation arises when we observe the values but **do not observe the labels**.

Now hide the labels

Suppose the same observations are placed together on one common axis.

At first, imagine that the colours reveal which component generated each observation.

```
x_all <- c(x1, x2)
z_all <- factor(
  c(rep("Component 1", length(x1)),
    rep("Component 2", length(x2)))
)

mixed_data <- data.frame(
  x = x_all,
  component = z_all
)
```

```r
ggplot(mixed_data, aes(x = x, y = 0, colour = component)) +
  geom_jitter(height = 0.08, width = 0, size = 2.5) +
  labs(
    title = "The labels are visible",
    x = "Observed value",
    y = NULL,
    colour = "Component"
  ) +
  scale_y_continuous(breaks = NULL) +
  theme_bw()
```

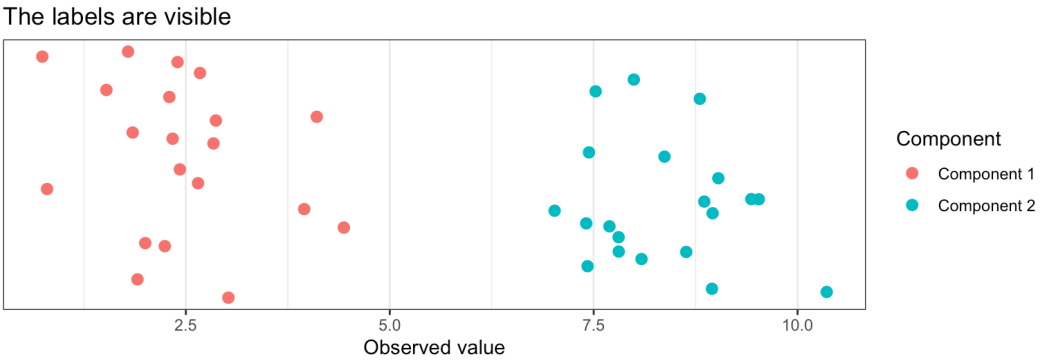


Figure 2: The observations are mixed together, but their component labels are still visible through colour.

Now imagine that we remove the colours.

```r
ggplot(mixed_data, aes(x = x, y = 0)) +
  geom_jitter(height = 0.08, width = 0, size = 2.5) +
  labs(
    title = "The labels are hidden",
    x = "Observed value",
    y = NULL
  ) +
  scale_y_continuous(breaks = NULL) +
  theme_bw()
```

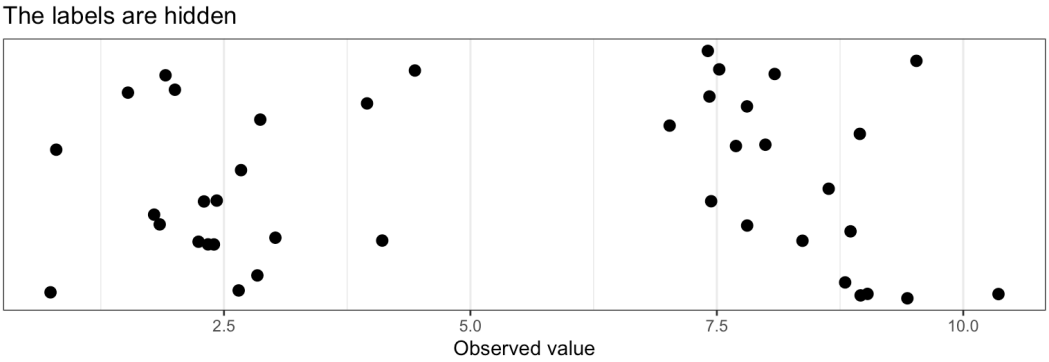


Figure 3: The observed data after the component labels have been hidden.

We still observe the values $X_1, \ldots, X_n$, but we no longer know which component generated each one.

For each observation, we introduce a variable

$$Z_i \in \{1, \ldots, K\},$$

where $Z_i = k$ means that observation i came from component k.

When Z_i is not observed, it is called a **latent variable**.

A useful way to think about this is:

> Z_i is the hidden label, or hidden colour, of observation i.

The generative process of a mixture model

A mixture model can be understood as a **two-stage random process**.

For each observation i:

1. First choose which component the observation comes from.
2. Then generate the observation from the distribution associated with that component.

Step 1: choose a component

Suppose there are K components. We write

$$Z_i \sim \text{Categorical}(\pi_1, \ldots, \pi_K),$$

where

$$\pi_k > 0, \qquad \sum_{k=1}^{K} \pi_k = 1.$$

Equivalently,

$$P(Z_i = k) = \pi_k.$$

The values $\pi_1, \ldots, \pi_K$ are called the **mixture proportions** or **mixture weights**.

They describe how common the different components are in the population.

For example, if

$$\pi_1 = 0.7, \qquad \pi_2 = 0.3,$$

then an observation has a 70% chance of being generated from component 1 and a 30% chance of being generated from component 2.

Step 2: generate the observation

Once component k has been selected, we generate the observation from the distribution associated with that component:

$$X_i \mid Z_i = k \sim P(X_i \mid \Theta_k), \qquad k = 1, \ldots, K.$$

For a Gaussian mixture model,

$$X_i \mid Z_i = k \sim N(\mu_k, \sigma_k^2).$$

So the main idea is:

> First choose **one** component. Then draw the observation from **that** component.

From the generative story to the mixture density

Suppose we observe $X_i = x_i$, but we do not observe Z_i.

The observation could have arisen from any of the K components. Using the law of total probability,

$$
\begin{aligned}
P(X_i) &= \sum_{k=1}^{K} P(X_i, Z_i = k) \\
&= \sum_{k=1}^{K} P(X_i \mid Z_i = k) P(Z_i = k) \\
&= \sum_{k=1}^{K} \pi_k P(X_i \mid \Theta_k).
\end{aligned}
$$

For a Gaussian mixture model, the density is therefore

$$
f(x_i) = \sum_{k=1}^{K} \pi_k f(x_i; \mu_k, \sigma_k^2).
$$

This is a **weighted sum of the component densities**.

The mixture weights appear because there are several possible ways in which an observation could have been generated.

Example 1: price of a randomly chosen book

Suppose a randomly selected book can be either paperback or hardback.

Assume:

- there is a 50% chance of selecting a paperback and a 50% chance of selecting a hardback;
- paperback prices follow $N(9, 1)$;
- hardback prices follow $N(20, 1)$.

For each book we can simulate the price in two steps:

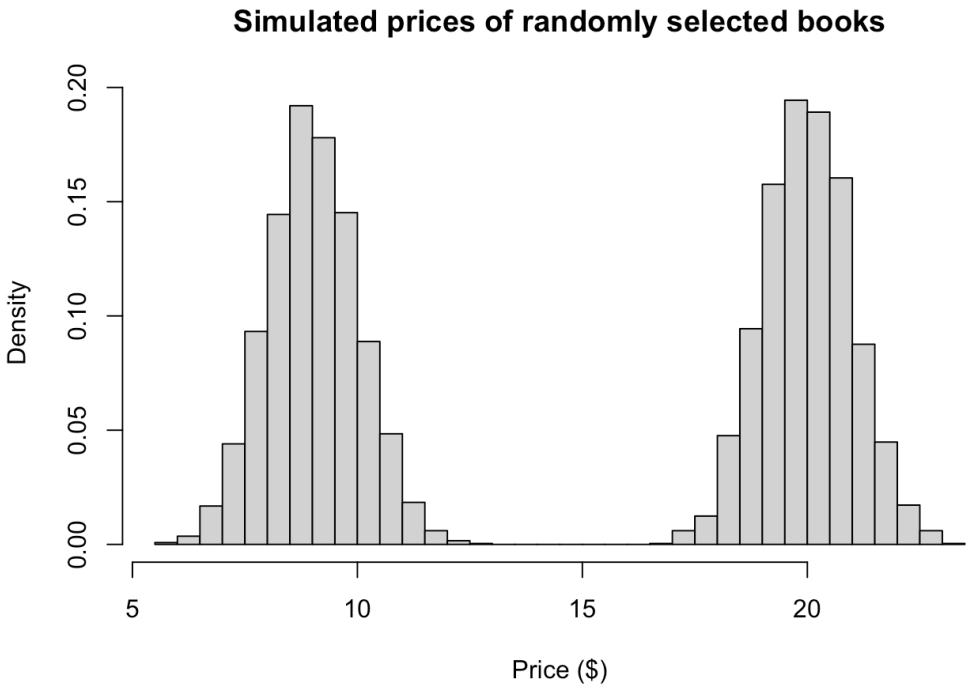
1. Draw the book type.
2. Conditional on the selected type, draw its price.

```
set.seed(30027)
NUM.SAMPLES <- 5000
prices <- rep(NA, NUM.SAMPLES)

for (i in seq_len(NUM.SAMPLES)) {
  z.i <- rbinom(1, 1, 0.5)
```

```
  if (z.i == 0) {
    prices[i] <- rnorm(1, mean = 9, sd = 1)
  } else {
    prices[i] <- rnorm(1, mean = 20, sd = 1)
  }
}

hist(
  prices,
  breaks = 35,
  probability = TRUE,
  xlab = "Price ($)",
  main = "Simulated prices of randomly selected books"
)
```



The histogram is clearly bimodal. Although each component is normal, the overall distribution of book prices is **not** normal.

To see why, we can draw the two component densities together with the mixture density.

```
mu.pb <- 9
sd.pb <- 1
mu.hb <- 20
sd.hb <- 1

sample.pts <- seq(5, 25, by = 0.1)
density_pb <- dnorm(sample.pts, mean = mu.pb, sd = sd.pb)
density_hb <- dnorm(sample.pts, mean = mu.hb, sd = sd.hb)
mixture_density_books <- 0.5 * density_pb + 0.5 * density_hb

plot(
  sample.pts,
  density_pb,
  col = "red",
  type = "l",
```

```
  xlab = "Price ($)",
  ylab = "Density",
  lty = 2,
  ylim = range(c(density_pb, density_hb, mixture_density_books))
)
lines(sample.pts, density_hb, col = "blue", lty = 2)
lines(sample.pts, mixture_density_books, col = "black", lwd = 2)

legend(
  "topright",
  c("paperback", "hardback", "all books"),
  col = c("red", "blue", "black"),
  lty = c(2, 2, 1),
  lwd = c(1, 1, 2),
  cex = 0.8
)
```

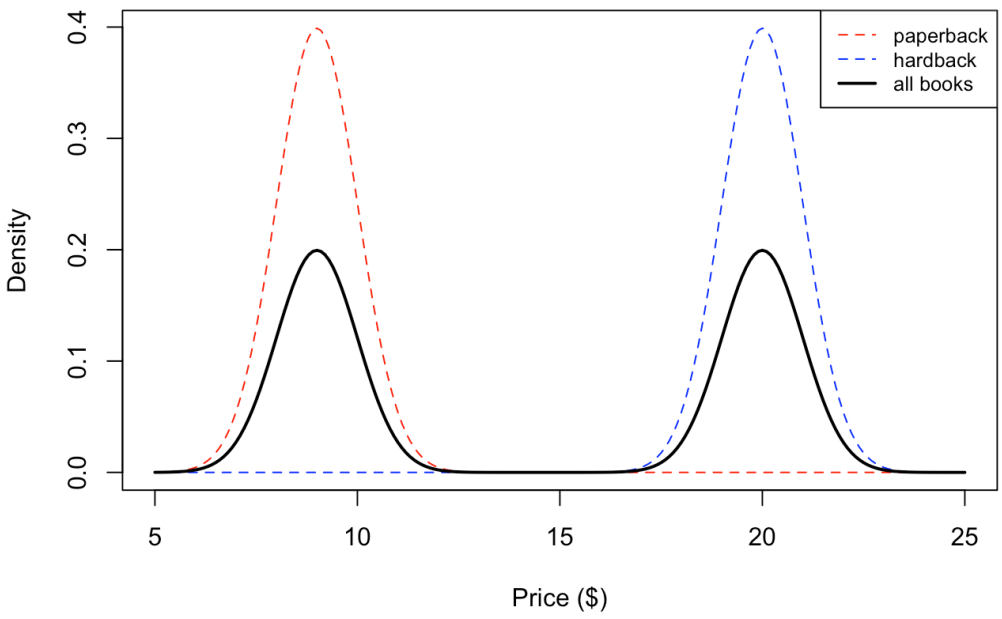


Figure 4: Paperback and hardback component densities and their 50/50 mixture.

For this model,

$$f(x) = 0.5 f_{\text{paperback}}(x) + 0.5 f_{\text{hardback}}(x).$$

A mixture does not need to be bimodal

A common first impression is that mixture models must produce distributions with several obvious peaks. This is **not** true.

Whether the mixture is visibly multimodal depends on factors such as:

- how far apart the component means are;
- the variability within each component; and
- the mixture proportions.

If the component distributions overlap strongly, the resulting mixture can be unimodal.

Example 2: heights of students

Suppose we model student heights using two component distributions:

$$X \mid Z = \text{male} \sim N(175, 6^2),$$

and

$$X \mid Z = \text{female} \sim N(160, 6^2).$$

Assume 25% of the population is male and 75% is female.

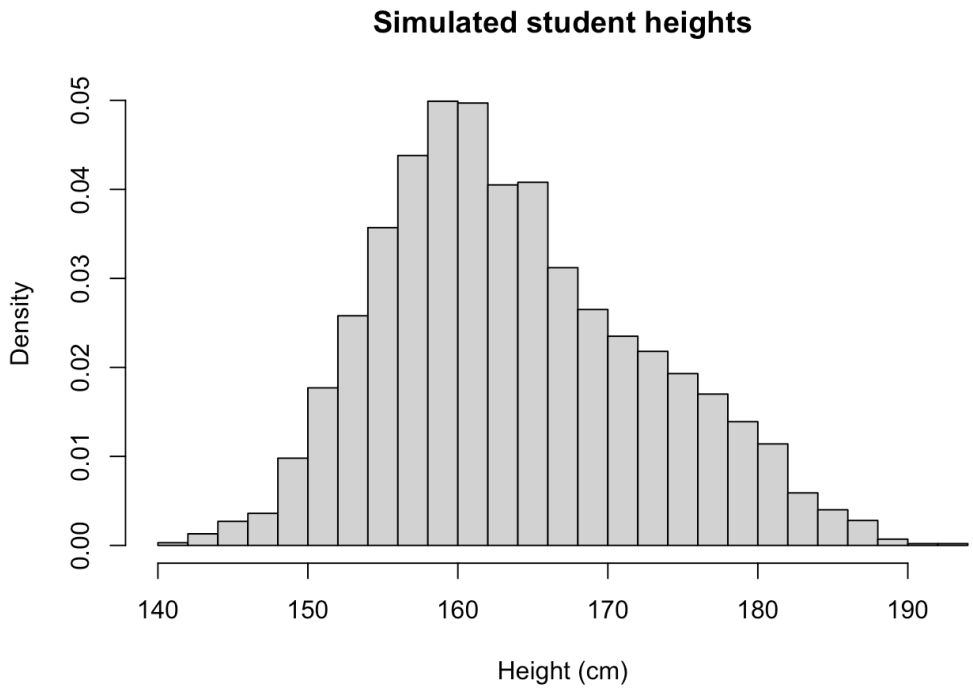
In the code below we use $Z = 0$ for male and $Z = 1$ for female, so

$$P(Z = 0) = 0.25, \qquad P(Z = 1) = 0.75.$$

```r
set.seed(30027)
NUM.SAMPLES <- 5000
heights <- rep(NA, NUM.SAMPLES)

for (i in seq_len(NUM.SAMPLES)) {
  z.i <- rbinom(1, 1, 0.75)
  if (z.i == 0) {
    heights[i] <- rnorm(1, mean = 175, sd = 6)
  } else {
    heights[i] <- rnorm(1, mean = 160, sd = 6)
  }
}

hist(
  heights,
  breaks = 35,
  probability = TRUE,
  xlab = "Height (cm)",
  main = "Simulated student heights"
)
```



Here the histogram is unimodal, even though the observations are generated from a two-component Gaussian mixture model.

```
mu.male <- 175
sd.male <- 6
mu.female <- 160
sd.female <- 6

sample.pts <- seq(130, 200, by = 0.5)
density_male <- dnorm(sample.pts, mean = mu.male, sd = sd.male)
density_female <- dnorm(sample.pts, mean = mu.female, sd = sd.female
mixture_density_heights <- 0.25 * density_male + 0.75 * density_fema

plot(
  sample.pts,
  density_male,
  col = "red",
  type = "l",
  xlab = "Height (cm)",
  ylab = "Density",
  lty = 2,
  ylim = range(c(density_male, density_female, mixture_density_heigh
)
lines(sample.pts, density_female, col = "blue", lty = 2)
lines(sample.pts, mixture_density_heights, col = "black", lwd = 2)

legend(
  "topright",
  c("male", "female", "population"),
  col = c("red", "blue", "black"),
  lty = c(2, 2, 1),
  lwd = c(1, 1, 2),
  cex = 0.8
)
```

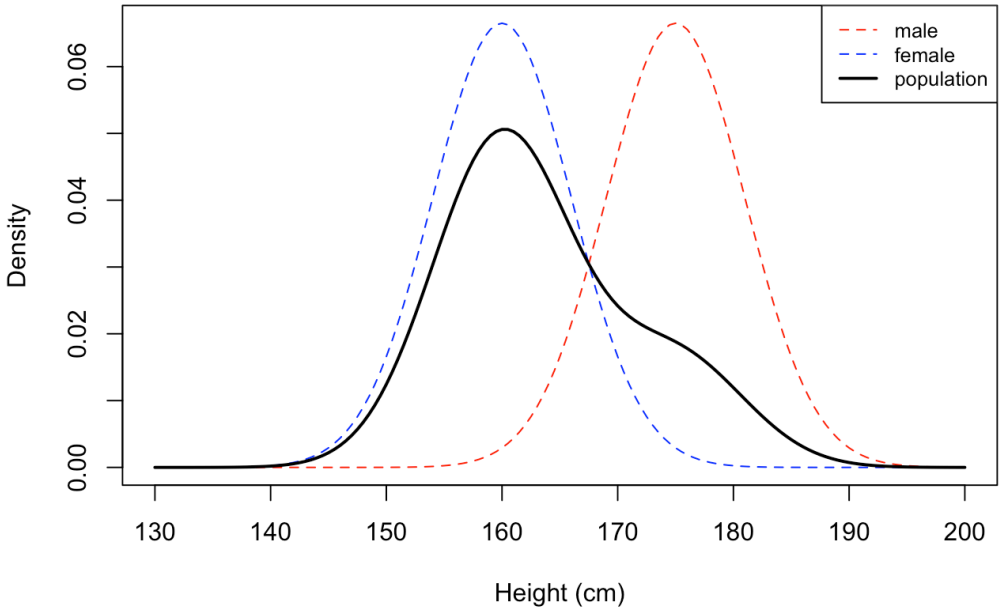


Figure 5: Male and female component densities and the resulting population mixture density.

The mixture is unimodal because the two component distributions overlap substantially. The population density is nevertheless not normal: it is asymmetric and is formed from a weighted combination of the two normal component densities.

Mixture model versus linear combination

There is an important distinction between a **mixture of distributions** and a **linear combination of random variables**.

A mixture model generates an observation by:

1. selecting one component at random; and
2. drawing the observation from that selected component.

It does **not** generate the observation by drawing from all components and then averaging the resulting values.

For example,

$$X = \pi_1 X_1 + \pi_2 X_2$$

is a linear combination of random variables. It is not the same construction as a mixture model.

The key distinction is:

> **Mixture:** choose one component, then sample from it.
> **Linear combination:** sample the random variables and combine their numerical values.

Clustering inference: which component generated an observation?

So far we have described how observations are generated. We now reverse the question.

Suppose we observe a value $X_i = x_i$. Which component is it likely to have come from?

More precisely, we want to compute

$$P(Z_i = k \mid X_i = x_i).$$

Using Bayes' rule,

$$P(Z_i = k \mid X_i) = \frac{P(X_i \mid Z_i = k)P(Z_i = k)}{\sum_{k'=1}^{K} P(X_i \mid Z_i = k')P(Z_i = k')}.$$

For a mixture model this becomes

$$P(Z_i = k \mid X_i) = \frac{\pi_k f_k(X_i)}{\sum_{k'=1}^{K} \pi_{k'} f_{k'}(X_i)}.$$

These probabilities are sometimes called **posterior component probabilities** or **responsibilities**.

The important point is that clustering is naturally **probabilistic**. If an observation lies in a region where two component distributions overlap, we may be uncertain about which component generated it.

Book example: posterior component probabilities

We now simulate 100 book prices and retain the true book type only so that we can compare our inferred probabilities with the truth.

```
set.seed(30027)
NUM.SAMPLES <- 100
X <- rep(NA, NUM.SAMPLES)
Z <- rep(NA, NUM.SAMPLES)

for (i in seq_len(NUM.SAMPLES)) {
  Z[i] <- rbinom(1, 1, 0.5)
  if (Z[i] == 0) {
    X[i] <- rnorm(1, mean = 9, sd = 1)
  } else {
    X[i] <- rnorm(1, mean = 20, sd = 1)
  }
}
```

Here:

- X contains the observed book prices;
- Z contains the true book types;
- $Z = 0$ denotes paperback;
- $Z = 1$ denotes hardback.

For the purpose of this example, pretend that we observe only X and not Z.

We also assume that the model parameters have already been estimated. To focus on the clustering calculation, we use the true parameter values:

$$\hat{\pi}_1 = 0.5, \qquad \hat{\mu}_1 = 9, \qquad \hat{\sigma}_1 = 1,$$

and

$$\hat{\pi}_2 = 0.5, \qquad \hat{\mu}_2 = 20, \qquad \hat{\sigma}_2 = 1.$$

In practice, these parameters would need to be estimated, for example using the EM algorithm.

Step 1: likelihood under each component

For each observation, calculate

$$P(X_i \mid Z_i = C_1) \qquad \text{and} \qquad P(X_i \mid Z_i = C_2).$$

```
L <- matrix(NA, nrow = length(X), ncol = 2)

# Component 1: paperback
L[, 1] <- dnorm(X, mean = 9, sd = 1)

# Component 2: hardback
L[, 2] <- dnorm(X, mean = 20, sd = 1)
```

Step 2: include the prior mixture proportions

We calculate the joint contributions

$$P(X_i, Z_i = C_k) = P(X_i \mid Z_i = C_k)P(Z_i = C_k).$$

```
M <- matrix(NA, nrow = length(X), ncol = 2)

M[, 1] <- L[, 1] * 0.5
M[, 2] <- L[, 2] * 0.5
```

Step 3: normalise to obtain component probabilities

```
CP <- M / rowSums(M)
```

Thus:

- `CP[, 1]` is $P(Z_i = C_1 \mid X_i)$;
- `CP[, 2]` is $P(Z_i = C_2 \mid X_i)$.

```
par(mfrow = c(2, 1))

par(mar = c(2, 4, 1, 1))
col_list <- c("red", "blue")[Z + 1]

plot(
  X,
  CP[, 2],
  col = col_list,
  xlim = c(5, 25),
  ylab = expression(P(Z[i] == C[2] ~ "|" ~ X[i])),
  pch = 19,
  cex = 0.7
)
legend(
  "bottomright",
  c("true: paperback", "true: hardback"),
  col = c("red", "blue"),
  pch = 19,
  cex = 0.7
```

```r
)

par(mar = c(4, 4, 1, 1))
plot(
  sample.pts <- seq(5, 25, by = 0.1),
  dnorm(sample.pts, mean = 9, sd = 1),
  col = "red",
  type = "l",
  xlab = "Price ($)",
  ylab = "Density",
  lty = 2,
  xlim = c(5, 25)
)
lines(sample.pts, dnorm(sample.pts, mean = 20, sd = 1), col = "blue"
lines(
  sample.pts,
  0.5 * dnorm(sample.pts, mean = 9, sd = 1) +
    0.5 * dnorm(sample.pts, mean = 20, sd = 1),
  col = "black",
  lwd = 2
)
legend(
  "topright",
  c("paperback", "hardback", "all books"),
  col = c("red", "blue", "black"),
  lty = c(2, 2, 1),
  lwd = c(1, 1, 2),
  cex = 0.7
)

par(mfrow = c(1, 1))
```

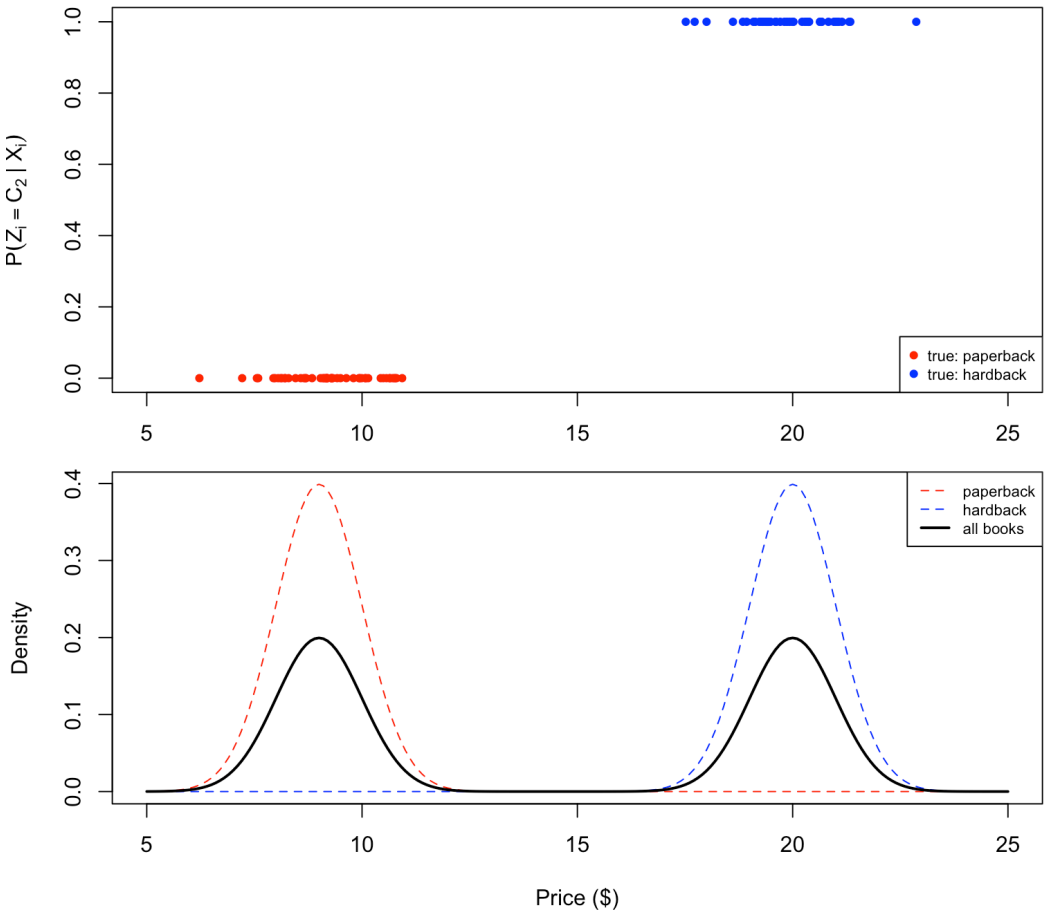


Figure 6: Posterior probability of belonging to the hardback component as a function of observed price.

Because the two book-price components are very well separated, most observations receive posterior probabilities very close to either 0 or 1.

Height example: posterior component probabilities

We repeat the same calculation for the student-height example.

```
set.seed(30027)
NUM.SAMPLES <- 100
X <- rep(NA, NUM.SAMPLES)
Z <- rep(NA, NUM.SAMPLES)

for (i in seq_len(NUM.SAMPLES)) {
  Z[i] <- rbinom(1, 1, 0.75)
  if (Z[i] == 0) {
    X[i] <- rnorm(1, mean = 175, sd = 6)
  } else {
    X[i] <- rnorm(1, mean = 160, sd = 6)
  }
}
```

Here $Z = 0$ denotes male and $Z = 1$ denotes female.

Again, pretend that only the heights X are observed.

We use

$$\hat{\pi}_{\text{male}} = 0.25, \qquad \hat{\mu}_{\text{male}} = 175, \qquad \hat{\sigma}_{\text{male}} = 6,$$

and

$$\hat{\pi}_{\text{female}} = 0.75, \qquad \hat{\mu}_{\text{female}} = 160, \qquad \hat{\sigma}_{\text{female}} = 6.$$

```
L <- matrix(NA, nrow = length(X), ncol = 2)

# Component 1: male
L[, 1] <- dnorm(X, mean = 175, sd = 6)

# Component 2: female
L[, 2] <- dnorm(X, mean = 160, sd = 6)

M <- matrix(NA, nrow = length(X), ncol = 2)
M[, 1] <- L[, 1] * 0.25
M[, 2] <- L[, 2] * 0.75

CP <- M / rowSums(M)
```

```
par(mfrow = c(2, 1))

par(mar = c(2, 4, 1, 1))
col_list <- c("red", "blue")[Z + 1]
```

```r
plot(
  X,
  CP[, 2],
  col = col_list,
  xlim = c(130, 200),
  ylab = expression(P(Z[i] == C[2] ~ "|" ~ X[i])),
  pch = 19,
  cex = 0.7
)
legend(
  "bottomright",
  c("true: male", "true: female"),
  col = c("red", "blue"),
  pch = 19,
  cex = 0.7
)

par(mar = c(4, 4, 1, 1))
sample.pts <- seq(130, 200, by = 0.5)
density_male <- dnorm(sample.pts, mean = 175, sd = 6)
density_female <- dnorm(sample.pts, mean = 160, sd = 6)

plot(
  sample.pts,
  density_male,
  col = "red",
  type = "l",
  xlab = "Height (cm)",
  ylab = "Density",
  lty = 2
)
lines(sample.pts, density_female, col = "blue", lty = 2)
lines(
  sample.pts,
  0.25 * density_male + 0.75 * density_female,
  col = "black",
  lwd = 2
)
legend(
  "topright",
  c("male", "female", "population"),
  col = c("red", "blue", "black"),
  lty = c(2, 2, 1),
  lwd = c(1, 1, 2),
  cex = 0.7
)

par(mfrow = c(1, 1))
```

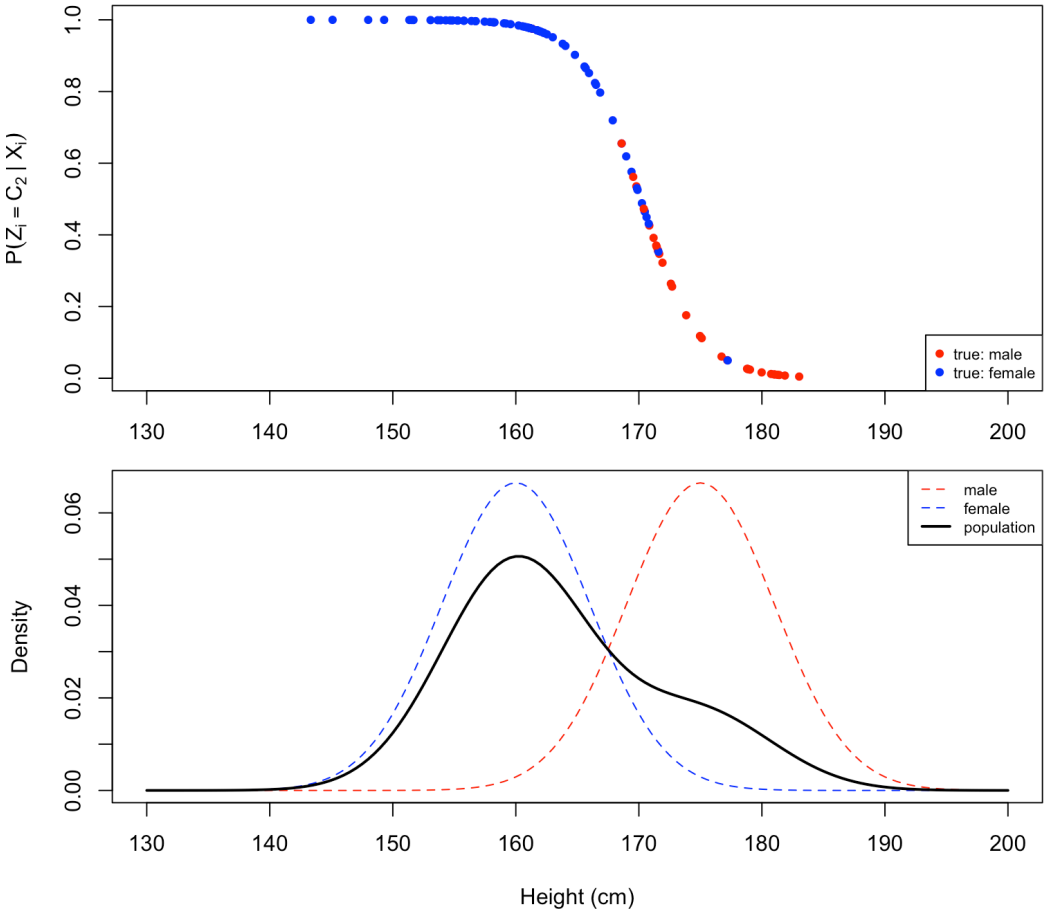


Figure 7: Posterior probability of belonging to the female component as a function of observed height.

Unlike the book example, there is substantial overlap between the two height distributions. Therefore, some observations have posterior component probabilities that are not close to 0 or 1.

This is exactly where the probabilistic nature of mixture-model clustering is useful: the model can represent **uncertainty about group membership** rather than forcing every observation into a group with complete certainty.

Summary

A mixture model can be understood through the following sequence:

1. The population contains several underlying components.
2. For each observation, a latent variable Z_i selects one component.
3. Conditional on that component, X_i is generated from the corresponding distribution.
4. If Z_i is not observed, the marginal distribution of X_i is a weighted mixture of the component distributions.
5. Given an observed value X_i, Bayes' rule allows us to calculate the probability that it came from each component.

The essential idea to remember is:

> **Choose one component, then generate the observation from that component.**